Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

June 6, 2024

Instructions: Read the questions carefully, show all your work, and explain all your answers. Indicate carefully if you introduce notation, or make assumptions when answering a question. Please make sure to allocate time to ALL three problems.

Good luck!

1. Consider an infinite-horizon discrete-time economy, which has access to a linear production technology $f(k) = Ak$, where $A > 0$ and k is the capital used in production. The output produced using this technology can be used for consumption or saved as capital for the next period. In the beginning of each period, a disaster may occur (e.g., a flood, an earthquake, or a war) with probability $q \in (0,1)$. The disaster wipes out some of the capital stock, and only fraction $\rho \in (0,1)$ of the capital stock survives. The representative agent's preferences are given by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0,1)$ is the time discount factor, c_t is the agent's consumption in period t, and and $u : \mathbb{R}_+ \to \mathbb{R}_+$ is a strictly increasing, strictly concave, and continuously-differentiable function, such that $\lim_{c \to 0} u'(c) = +\infty$. In period 0, the capital stock is $k_0 > 0$.

(a) Intuitively, how would you expect optimal savings to respond to an increase in the likelihood of the disaster (i.e., higher q) or an increase in the disaster's severity (i.e., lower ρ)? Simply briefly outline the economic forces that may impact the optimal savings.

(b) Set up recursively the social planner's decision problem in this economy.

(c) Derive the Euler Equation and interpret it intuitively.

(d) Suppose that $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, $\sigma > 0, \sigma \neq 1$. Derive the formula for the optimal savings rule in terms of the model's parameters.

(e) How is the optimal savings rule affected by an increase in q? How is it affected by a decrease in ρ? Are these effects consistent with the intuition you provided in part (a) above?

2. Consider the following version of the overlapping generations model with capital. Time is discrete ($t = 0, 1, 2...$). Individuals live for two periods. Population grows at the constant rate n.

Individuals value two goods when young and old. A standard market good (c_1), that can be consumed and saved at young age, and a non-storable, non-tradable *home* good (c_2), that is produced using an individual's own labor only—think of cooking, cleaning and gardening. Individuals only supply labor services in the market when young. The time endowment of individuals in both periods is normalized to 1.

An individual born at time t chooses consumption of both goods at t and $t+1$, savings and hours worked in the market at home when young to maximize

$$U(c_{1,t}, c_{2,t}, c_{1,t+1}, c_{2,t+1}) = v(c_{1,t}, c_{2,t}) + \beta v(c_{1,t+1}, c_{2,t+1}),$$

where $\beta > 0$ and

$$v(c_1, c_2) \equiv \log(c_1) + \frac{{c_2}^{1-\sigma}}{1-\sigma}, \quad \sigma > 0.$$

Competitive firms have access to a Cobb-Douglas technology that uses capital (K) and labor services (L), to produce market goods. Labor efficiency (X) grows at the rate g.

The home good is produced using a linear technology $c_{2,t} = Ah_t$, all t, where h_t denotes hours devoted to the production of the home good at t, and $A > 0$ is a productivity parameter.

(a) Formulate the problem of young individuals born at t. Derive the corresponding first-order conditions that characterize optimal individual choices. Are work hours in the market and at home constant over time?

(b) Derive the law of motion for the capital-labor ratio in the production of market goods.

Discuss whether this environment is consistent with a balanced growth path in which market hours are constant.

(c) Suppose now that at time $t = t_0 - 1$ the economy is in the steady state. At the start of t_0, the productivity of labor in home goods production (A) goes up on a permanent basis. Describe the implied effects over time on the rate of return on capital.

Economist GV argues that hours worked in the market will remain constant since the economy is on a balanced-growth path at $t_0 - 1$. Is he right? Explain.

3. Consider an economy populated by a continuum of individuals of measure 1. Each individual maximizes

$$E \sum_{t=0}^{\infty} \beta^t u(c_{it}),$$

where $c_{it} \geq 0$ denotes consumption of individual i at time t, and $\beta \in (0, 1)$.

In each period, individuals are endowed with one unit of time and an "entrepreneurial idea" (described below), and choose whether to be a worker or an entrepreneur. A worker supplies labor services on a competitive labor market. An entrepreneur operates a production technology $zf(n)$, where the function f is a strictly increasing and strictly concave, and z is an idiosyncratic technological shock reflecting quality of an "entrepreneurial idea." Ideas follow a Markov process with the transition function Q on $[\underline{z}, \bar{z}]$, and are i.i.d. across individuals. Entrepreneurs hire labor services; their own time endowment is spent supervising the production process and thus is not included as part of the labor input in $f(\cdot)$. Entrepreneurs trade the consumption good that they produce on the competitive goods market.

Each time an individual decides to *switch* their status from being a worker to being an entrepreneur or vice versa, they incur a utility cost of $\gamma \geq 0$ in the period they switch. Individuals cannot save or borrow.

There is a government that has exogenous expenditures G_t that follows Markov process with the transition function P on $[\underline{G}, \bar{G}]$. The government runs a *balanced budget.* It finances its expenditures in each period by imposing a tax on profits, so that each entrepreneur has to pay a fraction τ of their profits to the government.

(a) Define a recursive competitive equilibrium in this economy.

(b) Assume that $\gamma = 0$, $G_t = G$ for all t, and ideas are i.i.d. over time. Suppose that the economy is in a stationary equilibrium, and G increases unexpectedly once and for all. What happens to the equilibrium wage rate? Explain.